THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A shiny solid X conducts electricity, is hard, and its oxide turns red litmus blue. X is probably: 1

(a) carbon **(b)** magnesium **(c)** sulphur **(d)** phosphorus

2. Which of these metals will give a vigorous reaction with dilute HCl? 1

(a) copper **(b)** gold **(c)** silver **(d)** magnesium

3. Which metal is used to coat iron sheets to protect them from rusting? 1

(a) coal **(b)** zinc **(c)** wax **(d)** sulphur

4. A yellow brittle solid Y burns with a blue flame and its oxide turns blue litmus red. Y is: 1

(a) copper **(b)** iron **(c)** magnesium **(d)** sulphur

5. In $\mathrm{Fe + CuSO_4 \longrightarrow FeSO_4 + Cu}$, the metal that displaces copper is: 1

(a) Fe **(b)** $\mathrm{FeSO_4}$ **(c)** Cu **(d)** $\mathrm{CuSO_4}$

6. Which of these metals is most suitable for making a very hot cooking pan? 1

(a) sulphur **(b)** wood **(c)** copper **(d)** lead

7. A metal Z reacts with dilute HCl giving hydrogen but does not react with cold water. Z may be: 1

(a) zinc **(b)** potassium **(c)** sodium **(d)** gold

8. In the reaction above, the colour of the solution changes from: 1

(a) blue to green **(b)** colourless to blue
(c) green to blue **(d)** blue to colourless

9. **Assertion (A):** Aluminium can be beaten into very thin foils. 1
Reason (R): Metals are malleable.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Graphite conducts electricity even though it is a non-metal. 1
Reason (R): Graphite is a form of carbon.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Predict the products when zinc reacts with dilute sulphuric acid and write the equation. 2

12. Why do non-metallic oxides cause acid rain? 2

13. Why do metals feel cold to the touch? 2

14. Explain why a gold ring does not need cleaning as often as a silver ring. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student has an unknown sample. It is shiny, ductile, burns in air to a white powder that turns red litmus blue. Identify the class and give a probable example. 3

16. Explain with equations how metals and non-metals form acidic and basic solutions with water. 3

17. Why can't we make hammers from sulphur or wax? 3

OR

Describe a test to distinguish a piece of copper from a piece of aluminium without chemicals. 3

18. Why is a small piece of sodium kept in kerosene reduced in size when cut? 3

OR

Give three differences between sodium and copper. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Aditi helps her mother in the kitchen in Pune. The frying pan is made of aluminium but its handle is made of plastic. Aditi also notices that aluminium foil can be wrapped around parathas, and that the copper wire of the mixer is thin and long. 5
(i) Why is the pan made of a metal?
(ii) Why is the handle made of plastic?
(iii) Which property of aluminium allows it to be made into foil?
(iv) Name the property that allows copper to be made into wire and one more property that makes copper suitable for electric wires.

20. Explain, with equations, the reactions of magnesium with oxygen, acid and copper sulphate solution. 5

OR

Discuss the use of metals for tools from the Bronze Age to the present. 5

END OF QUESTION PAPER — Total: 40 Marks